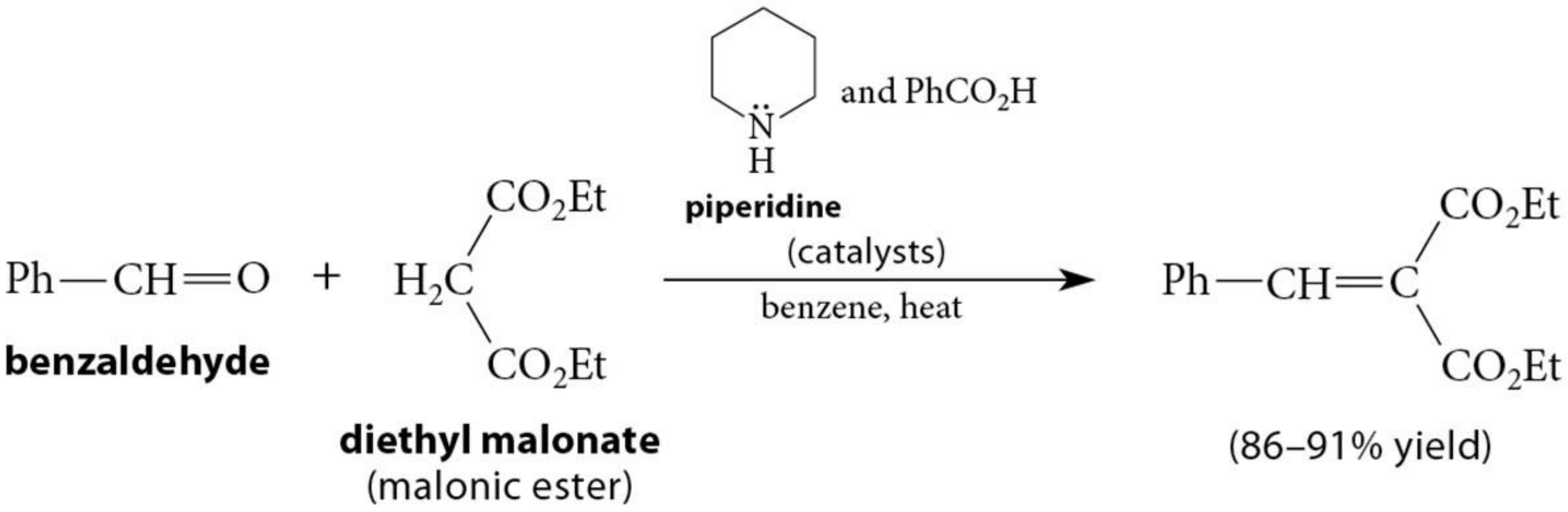
Chapter 22 (Carey Chapter 21)

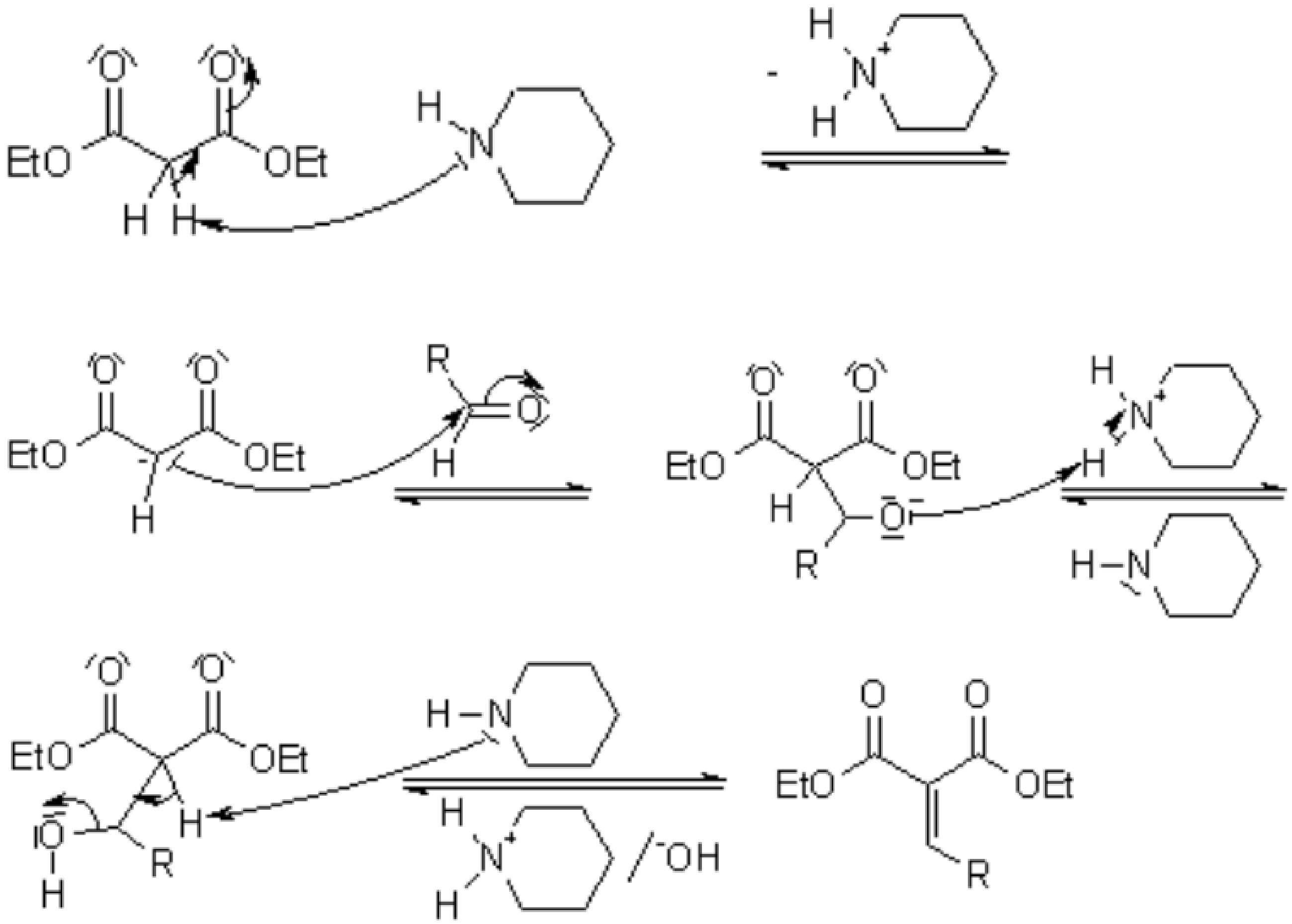
The Chemistry of Enolate Ions, Enols, and α,β-Unsaturated Carbonyl Compounds

Knoevenagel Condensation

- ***Knoevenagel Reaction*** - An aldol condensation reaction of malonic ester, acetoacetic ester, and other relatively acidic carbonyl compounds.



Knoevenagel Condensation

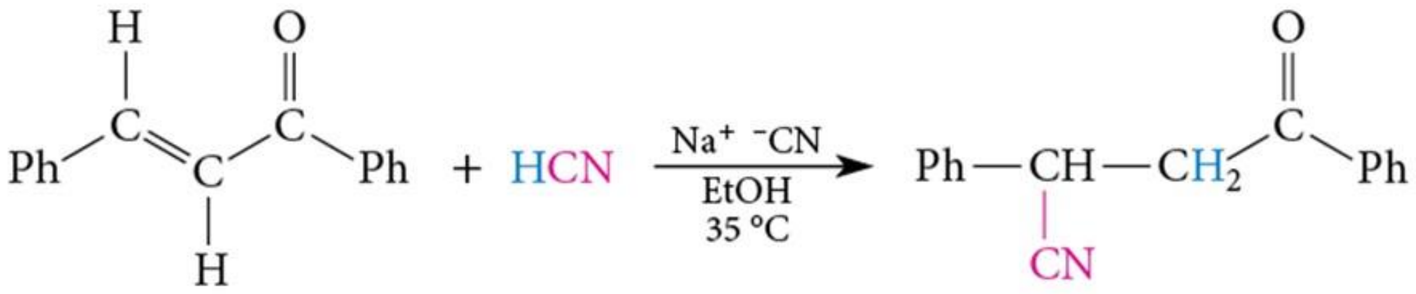


- ***Knoevenagel Reaction*** – is an aldol condensation reaction of malonic ester derivatives

https://www.organic-chemistry.org/namedreactions/knoevenagel-condensation.shtm

Conjugate Addition (Michael addition or 1,4-addition)

- A unique reactivity is observed for alkenes that are conjugated with carbonyls (C=C—C=O).
- These conjugated compounds are generally called α,β-unsaturated carbonyl compounds.



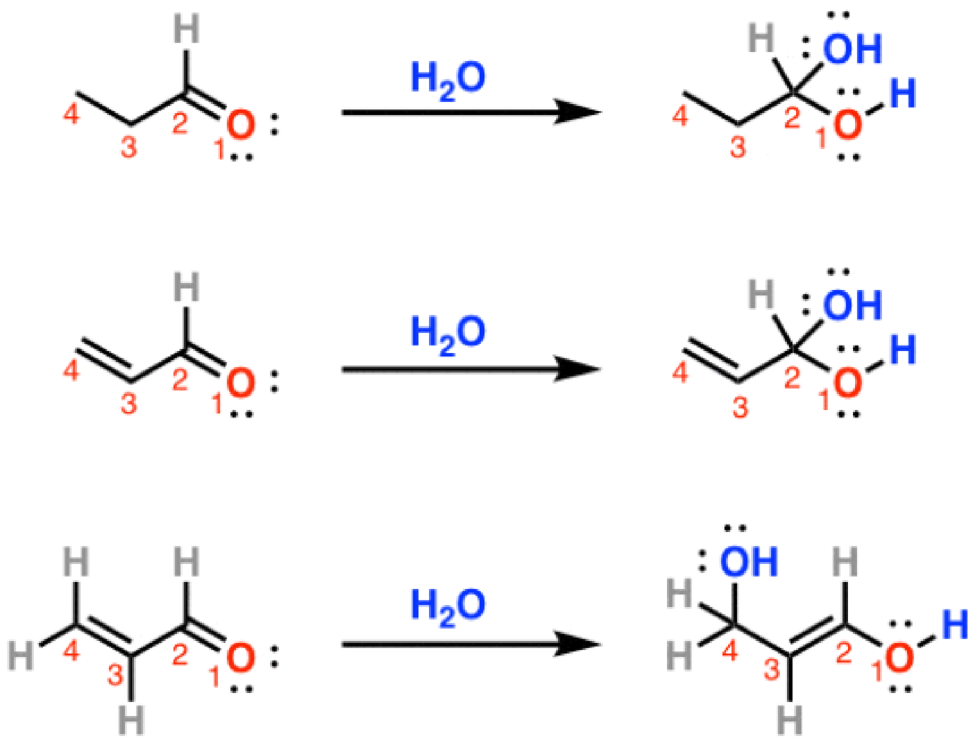
(trans isomer) (93–96% yield)

$$CH_3CH{=}CH_2 + HCN \xrightarrow[EtOH]{Na^+\ {}^-CN} \text{no reaction}$$

Conjugate Addition (Michael addition or 1,4-addition)

Types of addition reactions to carbonyls and α,β unsaturated carbonyls are named similarly, even though oxygen (1) is not part of the carbon chain

Example: addition of water



OH added to C_2
H added to O
"1,2-addition"
(Or just "addition")

OH added to C_2
H added to O
Called "1,2-addition"

OH added to C_4
H added to O
Called "1,4-addition"
or "conjugate addition"

22.9 Conjugate-Addition Reactions

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Conjugate Addition is Nucleophilic Addition

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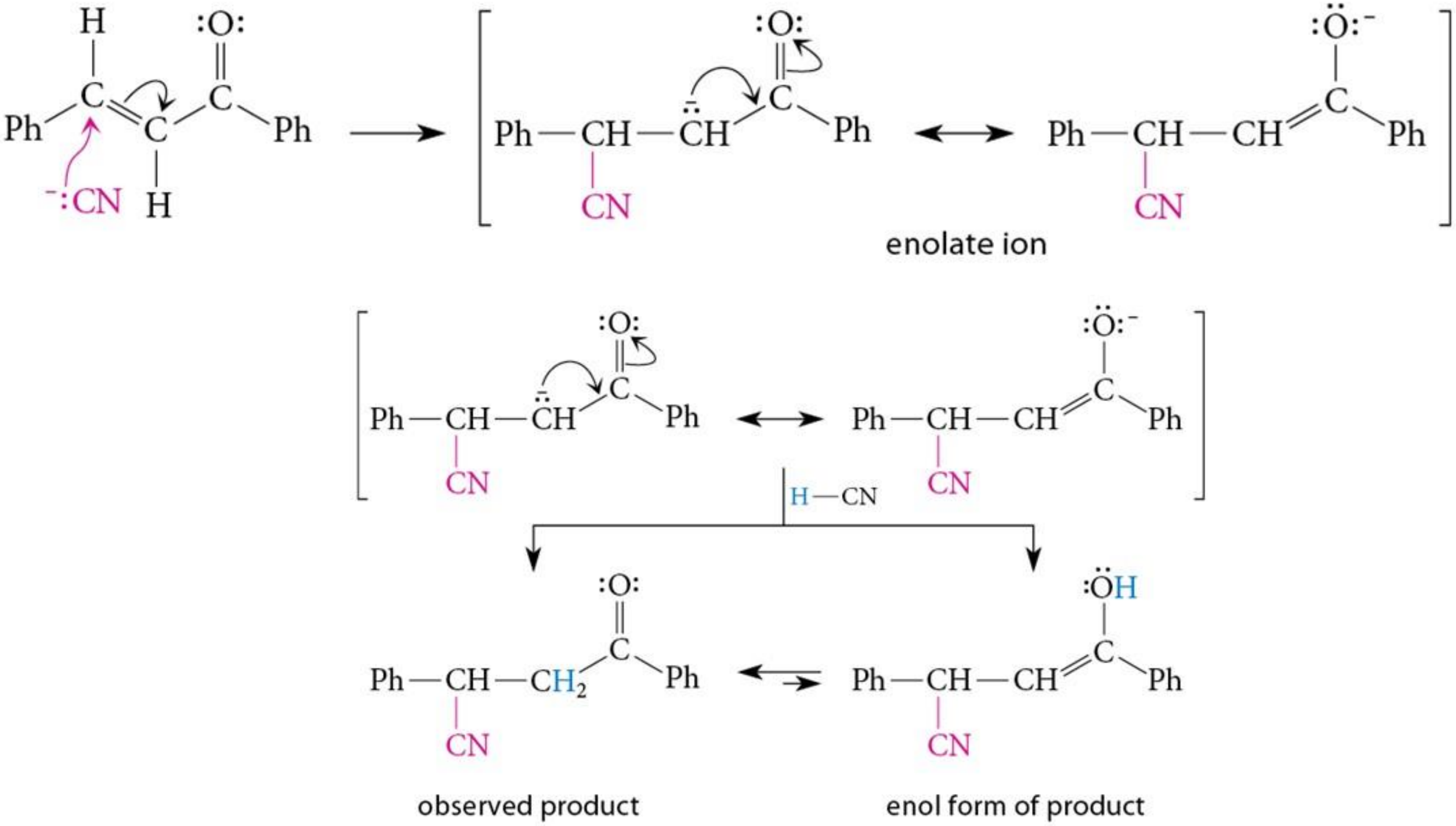
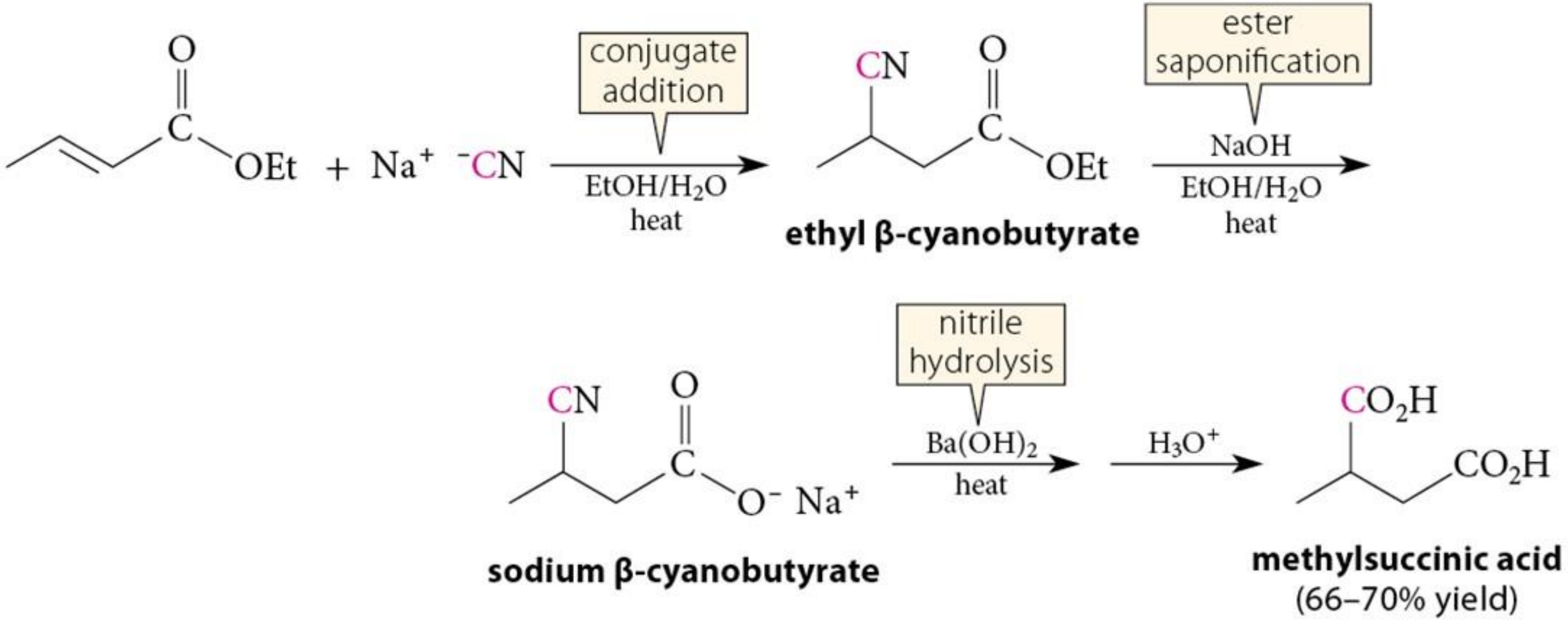
- Produces a resonance-stabilized enolate ion intermediate



Conjugate Addition With α,β-Unsaturated Esters

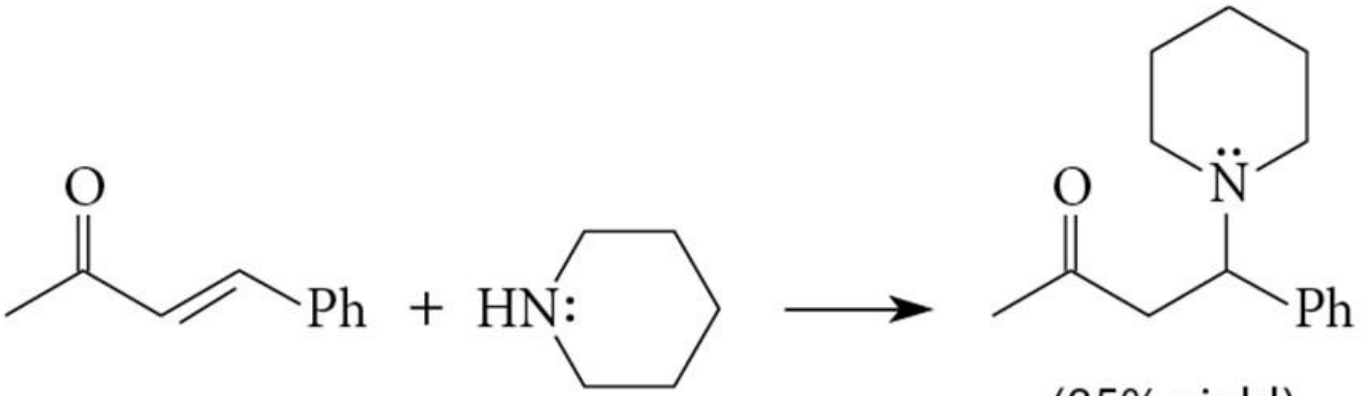


$$\underset{\textbf{2-propanethiol}}{(CH_3)_2CH{-}SH} + \underset{\textbf{methyl acrylate}}{H_2C{=}CH{-}CO_2Me} \xrightarrow[\text{MeOH}]{\text{NaOMe}} \underset{\textbf{methyl 3-(isopropylthio)propanoate}}{(CH_3)_2CH{-}S{-}CH_2{-}CH_2{-}CO_2Me}$$

methyl 3-(isopropylthio)propanoate (97% yield)

22.9 Conjugate-Addition Reactions

Conjugate Addition With α,β-Unsaturated Ketones and α,β-Unsaturated Nitriles



(85% yield)

a cyanoethylation

$$\underset{\textbf{methanethiol}}{CH_3SH} + \underset{\textbf{acrylonitrile}}{H_2C{=}CH{-}CN} \xrightarrow[\text{MeOH}]{\text{NaOMe}} CH_3S{-}\overbrace{CH_2{-}CH_2{-}CN}^{\text{a β-cyanoethyl group}}$$

3-(methylthio)propanenitrile
(91% yield)

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Acid-Catalyzed Conjugate Addition

- Acid catalyzed conjugate additions are also known.

$$H_2C{=}CH{-}CO_2Me + HBr \xrightarrow[Et_2O]{} Br{-}CH_2CH_2{-}CO_2Me$$

methyl acrylate

methyl β-bromopropionate
(80–84% yield)

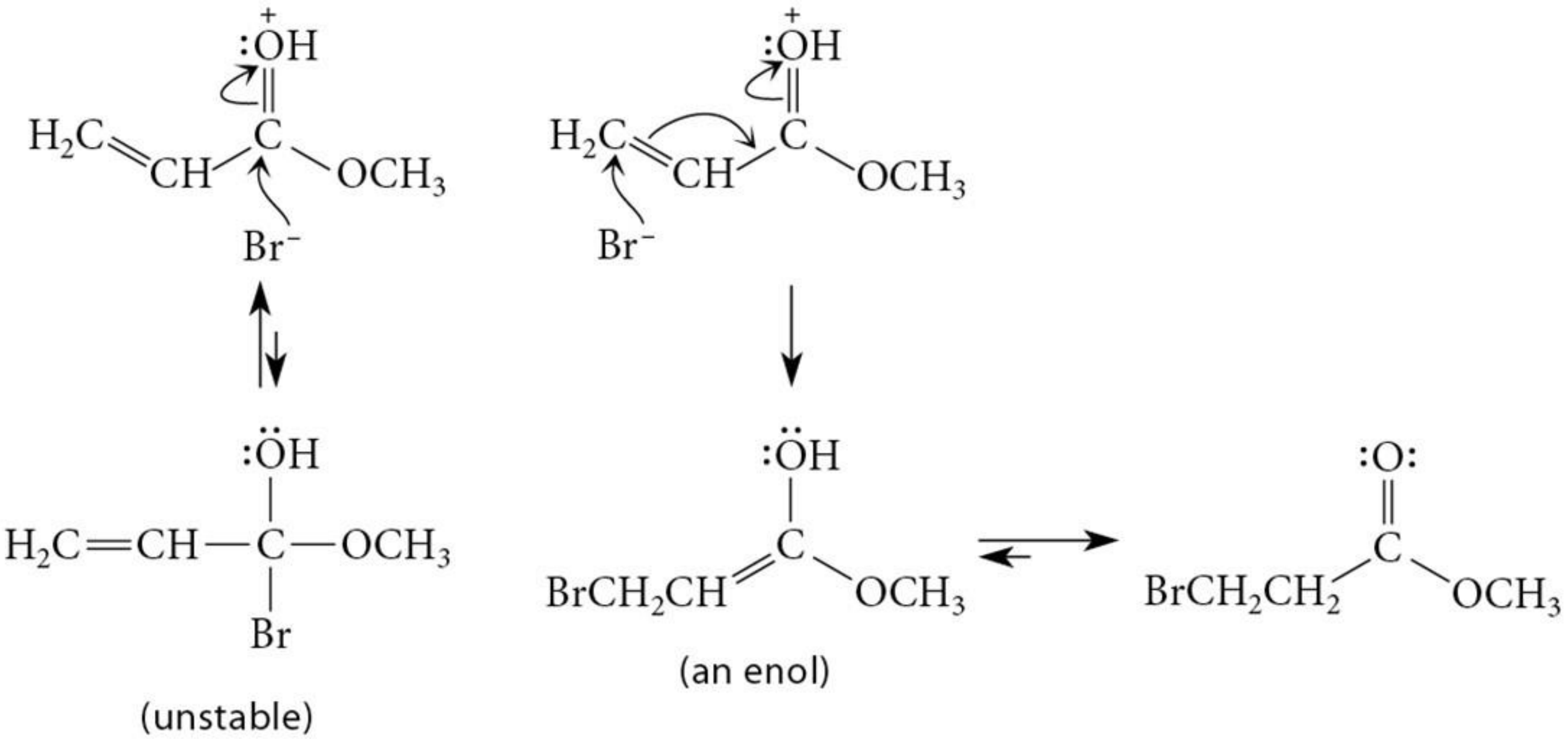
$$H_2C{=}CH{-}CH{=}O + HCl \xrightarrow[-15\,^{\circ}C]{} Cl{-}CH_2CH_2{-}CH{=}O$$

2-propenal
(acrolein)

3-chloropropanal

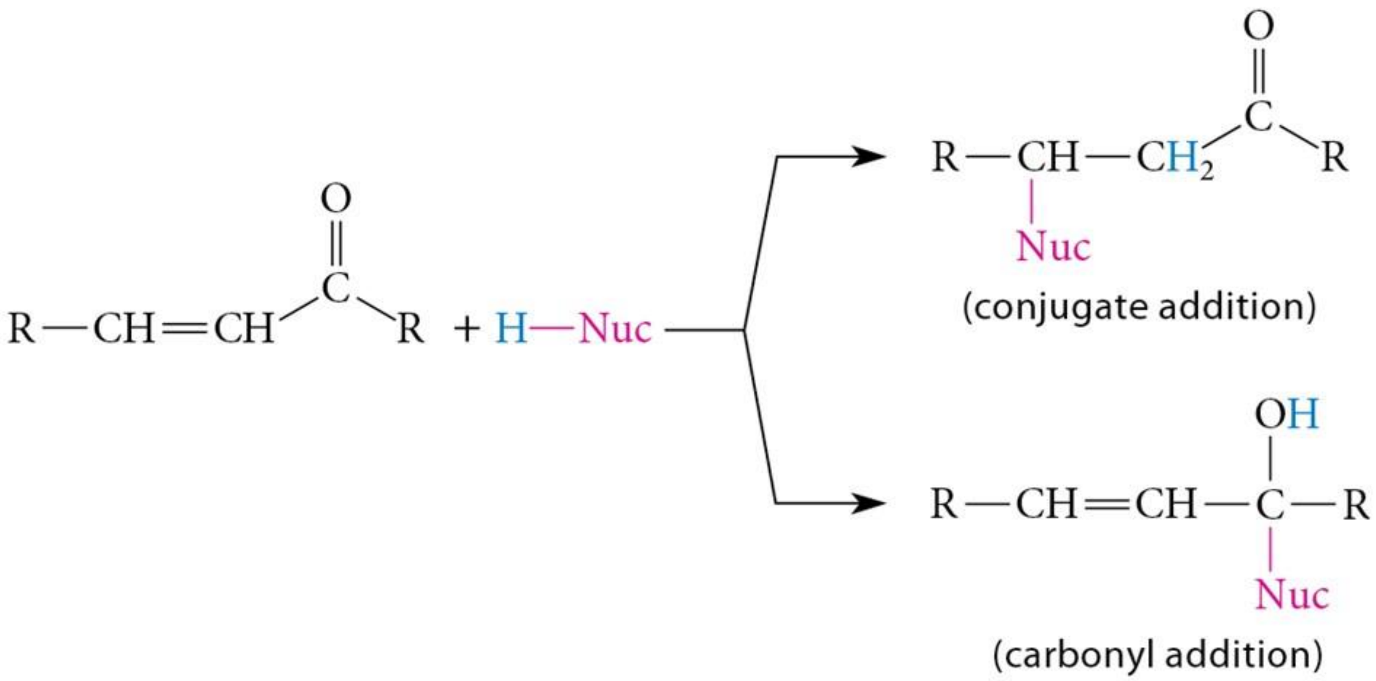
Mechanism of Acid-Catalyzed Conjugate Addition

- Carbonyl oxygen is protonated, not double bond C.



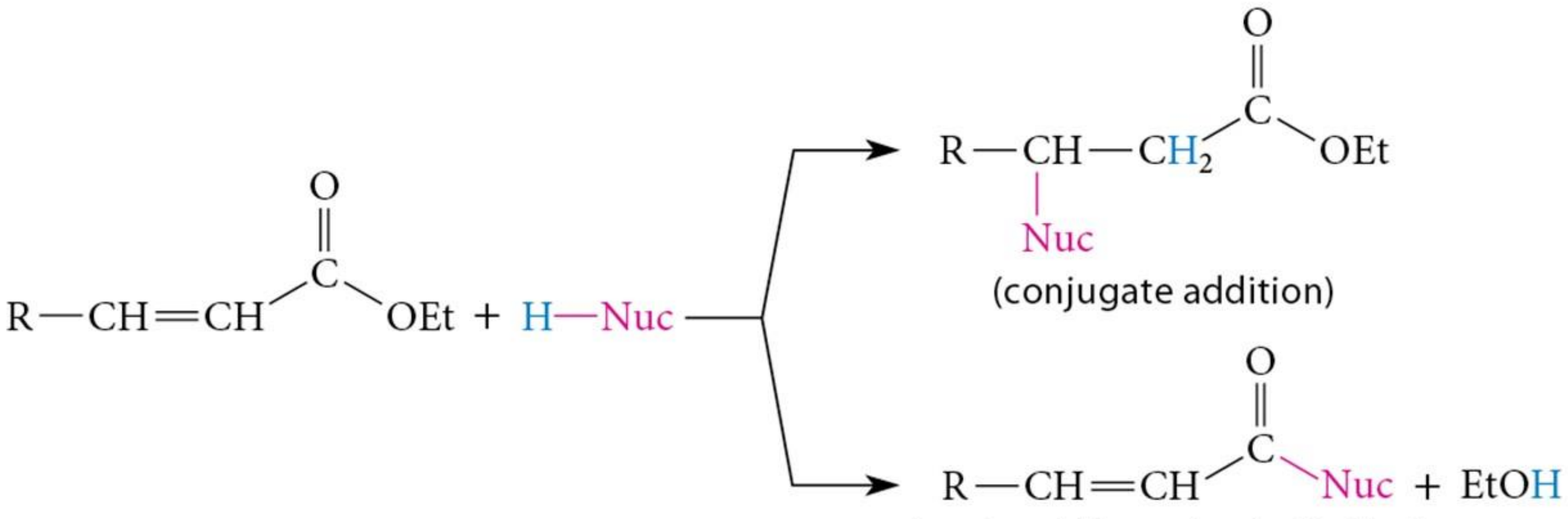
Conjugate Addition versus Carbonyl Addition

- For *ketones and aldehydes*, relatively weak bases that give *reversible* carbonyl-addition tend to give conjugate addition.

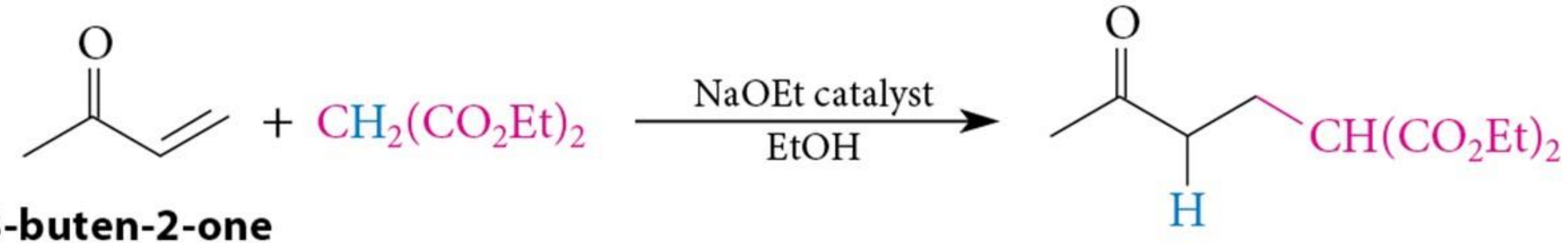


Conjugate Addition versus Nucleophilic Acyl Substitution

- For *esters* conjugate addition *competes* with nucleophilic acyl substitution.



Conjugate Addition of Enolate Ions

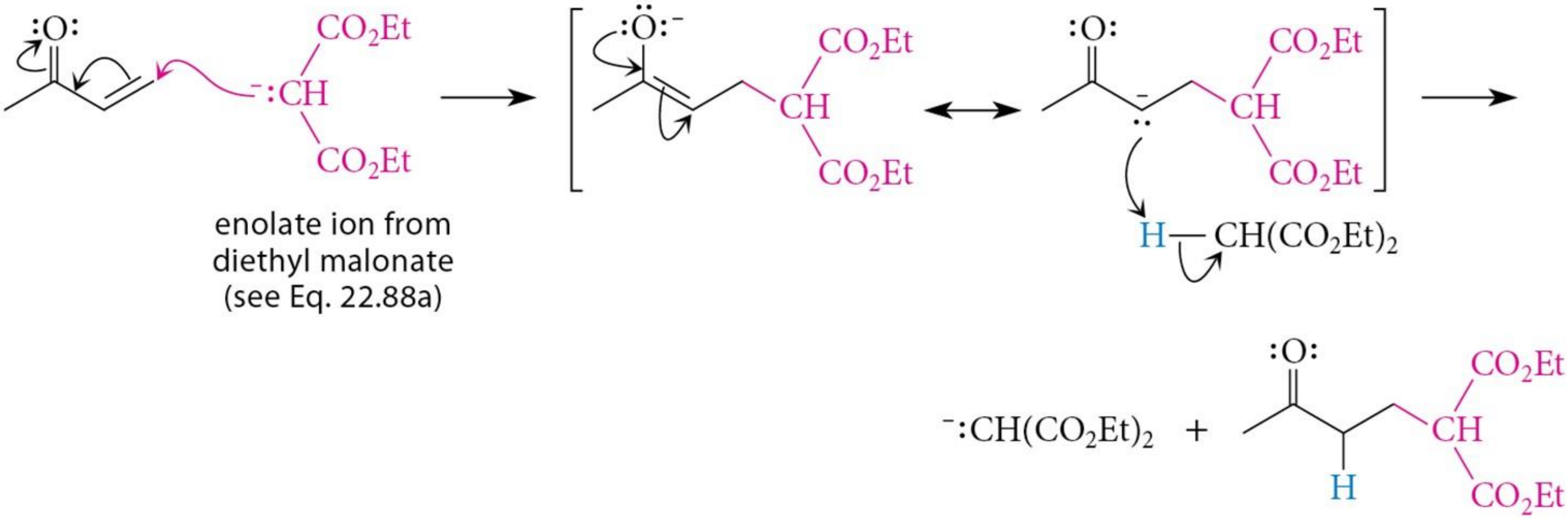


3-buten-2-one
(methyl vinyl ketone)

(65–71% yield)

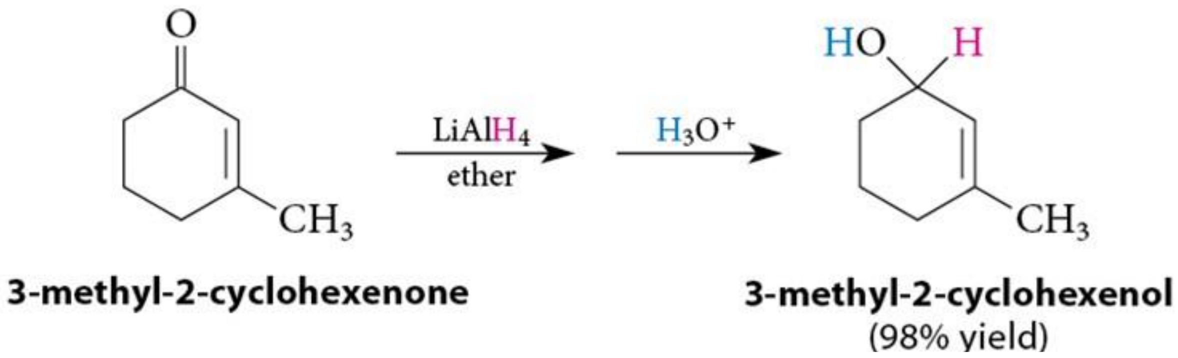
- Common for enolate ions derived from malonic ester derivatives and β-keto esters.
- Conjugate additions to α,β-unsaturated carbonyl compounds are called ***Michael additions.***

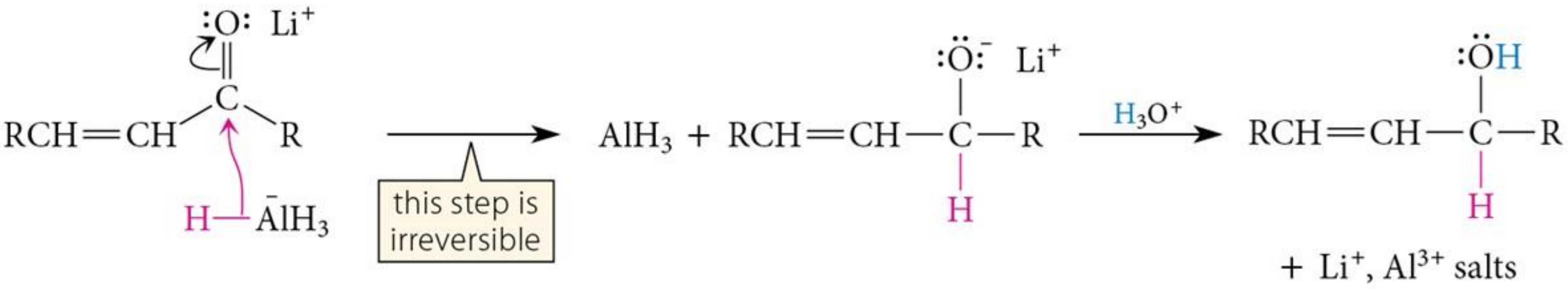
Michael Addition



Reduction of α,β-Unsaturated Carbonyls (1 of 2)

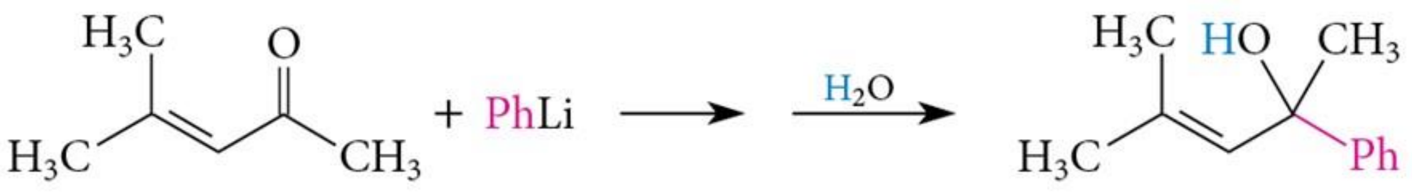
- Carbonyl reduction is not only *faster* than conjugate addition, but is also *irreversible* (i.e., *kinetically controlled*).





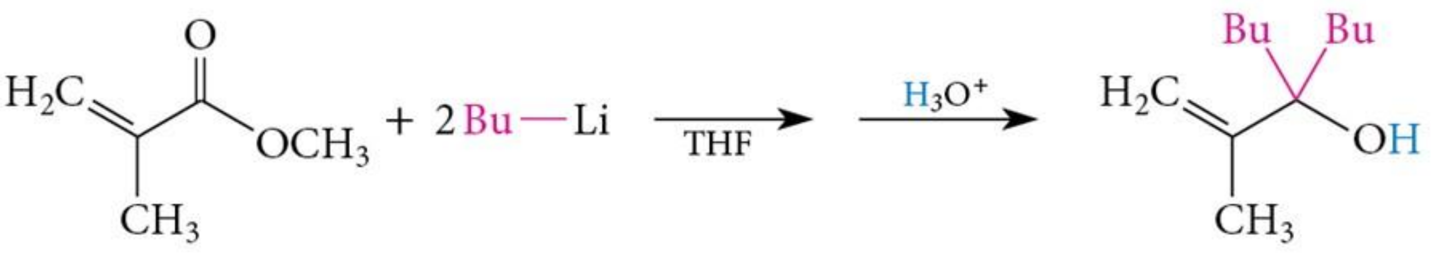
Addition of Organolithium Reagents to α,β-Unsaturated Carbonyls

- Addition of R—Li results in carbonyl addition.



4-methyl-3-penten-2-one
(mesityl oxide)

4-methyl-2-phenyl-3-penten-2-ol
(67% yield)

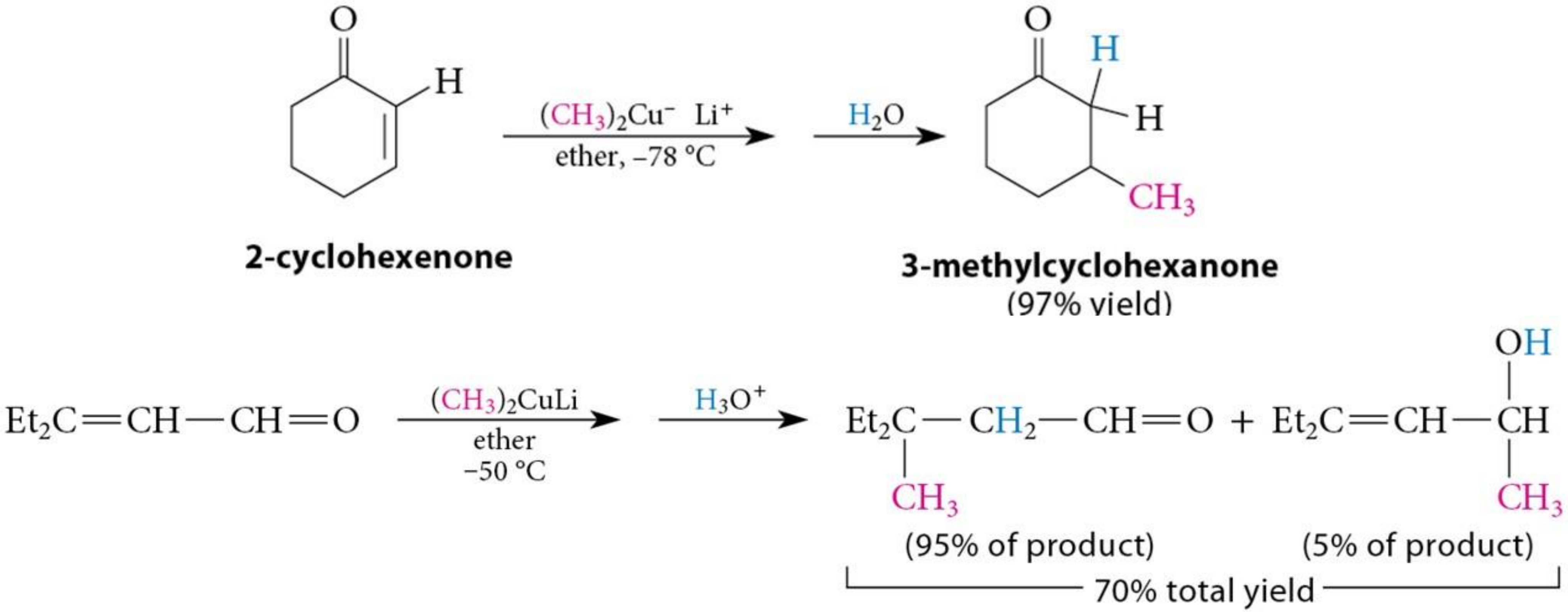


methyl
2-methylpropenoate

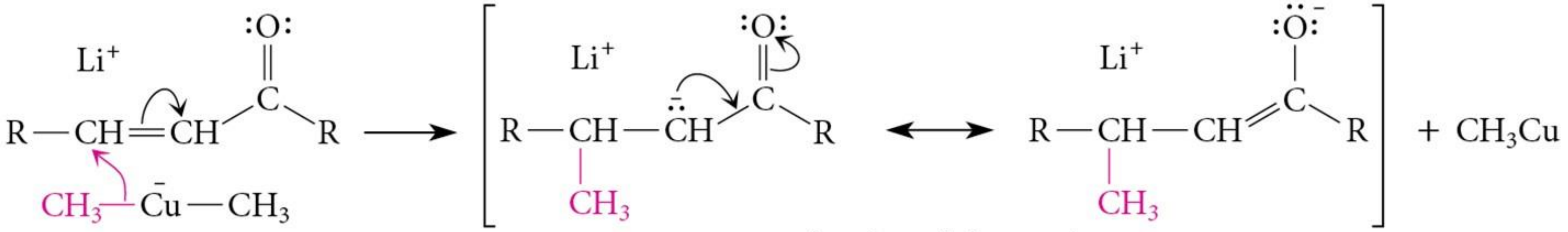
3-butyl-2-methyl-1-hepten-3-ol
(89% yield)

Addition of Lithium Dialkylcuprate Reagents to α,β-Unsaturated Carbonyls

- Results in conjugate addition.



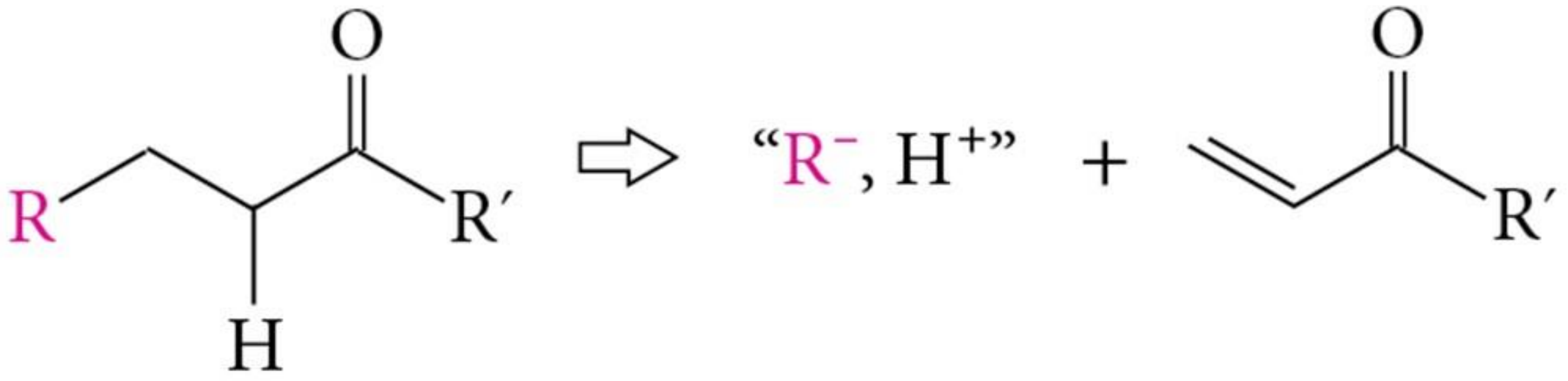
Possible Mechanism of Addition of Lithium Dialkylcuprate Reagents



enolate ion of the product

Conjugate Addition in Organic Synthesis

- Consider any group at the β-position of a carbonyl (or nitrile) as a nucleophile in a conjugate addition reaction.

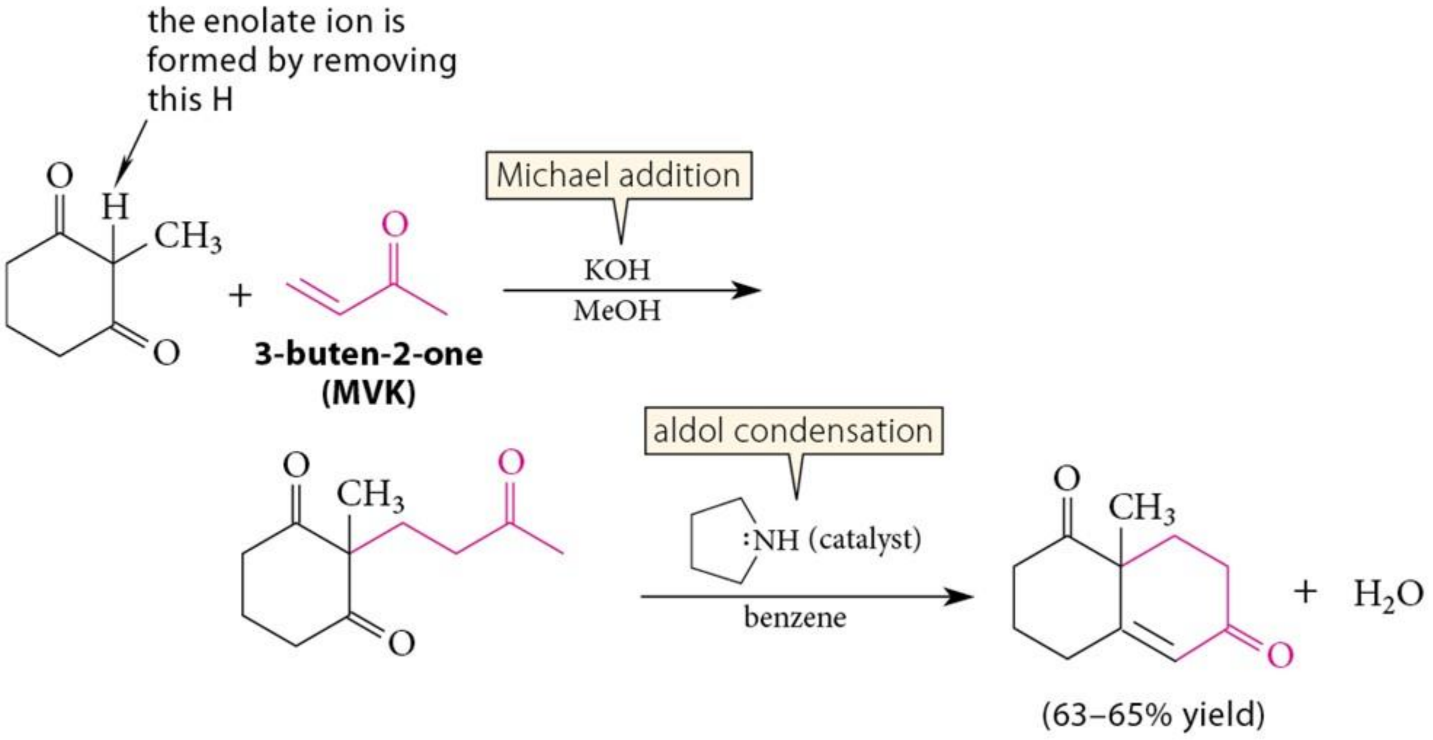


Summary of C—C Bond Forming Reactions

1. Aldol addition and condensation reactions
2. Claisen and Dieckmann condensations
3. Malonic ester synthesis
4. Acetoacetic ester synthesis
5. Alkylation of ester enolates
6. Aldol addition of ester enolates
7. Conjugate addition of cyanide ions and enolate ions
8. Conjugate addition of R_2CuLi

Optional study material: Robinson Annulation

- Michael addition followed by an aldol condensation that closes a ring.



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